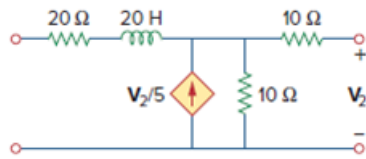


Problem

Three copies of the circuit shown in Fig. 19.70 are connected in cascade. Determine the z parameters.

Figure 19.70



Step-by-step solution

Step 1 of 12

Refer to Figure 19.70 from the textbook.

Sketch the circuit showing the two loops as shown in Figure 1.

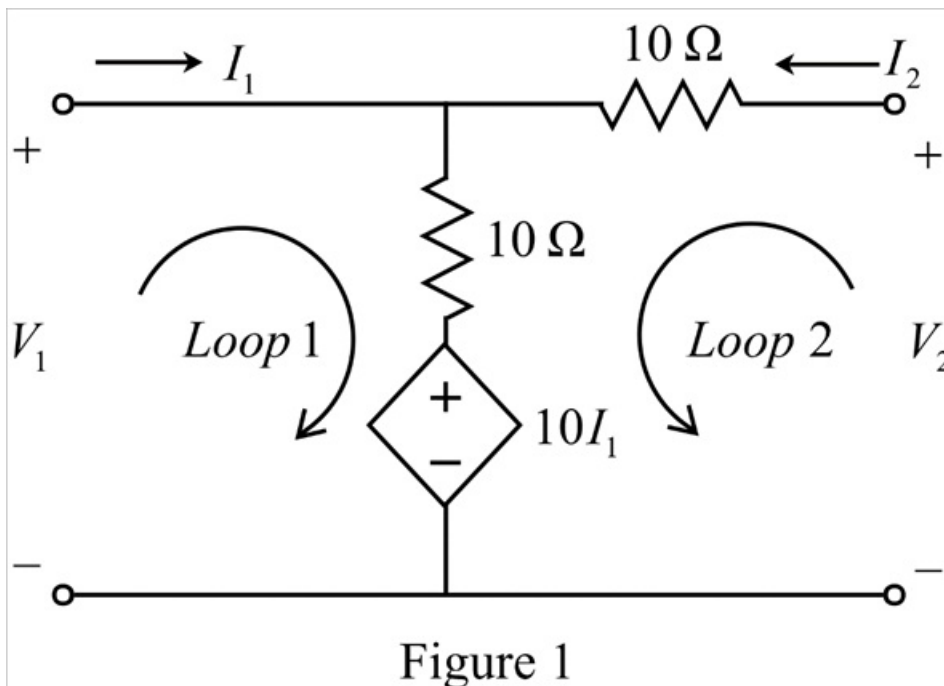
Here,

V_1 is the voltage across port 1,

V_2 is the voltage across port 2,

I_1 is the current entering into the network through port 1,

I_2 is the current entering into the network through port 2.



Step 2 of 12

Apply Kirchhoff's Voltage Law in Loop 1.

$$V_1 = 10(I_1 + I_2) + 10I_1$$

$$V_1 = 20I_1 + 10I_2 \dots\dots (1)$$

Apply Kirchhoff's Voltage Law in Loop 2.

$$V_2 = 10I_2 + 10(I_1 + I_2) + 10I_1$$

$$V_2 = 20I_1 + 20I_2$$

$$I_1 = \frac{V_2 - 20I_2}{20}$$

$$I_1 = 0.05V_2 - I_2 \dots\dots (2)$$

Step 3 of 12

Substitute $0.05V_2 - I_2$ for I_1 in equation (1)

$$V_1 = 20(0.05V_2 - I_2) + 10I_2$$

$$V_1 = V_2 - 20I_2 + 10I_2$$

$$V_1 = V_2 - 10I_2 \dots\dots (3)$$

Step 4 of 12

Represent equations (2) and (3) in matrix form as follows:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} 1 & 10 \\ 0.05 & 1 \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \dots\dots (4)$$

Write the equation for the *ABCD* parameters of a two-port network in matrix form.

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \dots\dots (5)$$

Compare equations (4) and (5) and find the *ABCD* parameters matrix.

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & 10 \\ 0.05 & 1 \end{bmatrix}$$

Thus, the *ABCD* parameter matrix, T_a is,

$$[T_a] = \begin{bmatrix} 1 & 10 \\ 0.05 & 1 \end{bmatrix}$$

Step 5 of 12

Since the overall $ABCD$ parameters of a cascaded network are the products of the $ABCD$ parameters of the individual networks.

Thus, the overall $ABCD$ parameters, T of cascaded network of three identical networks is,

$$[T] = [T_a][T_a][T_a]$$

$$\begin{bmatrix} A_o & B_o \\ C_o & D_o \end{bmatrix} = [T_a]^3$$

Substitute $\begin{bmatrix} 1 & 10 \\ 0.05 & 1 \end{bmatrix}$ for $[T_a]$.

$$\begin{aligned} [T] &= \begin{bmatrix} 1 & 10 \\ 0.05 & 1 \end{bmatrix}^3 \\ &= \begin{bmatrix} 1 & 10 \\ 0.05 & 1 \end{bmatrix} \begin{bmatrix} 1 & 10 \\ 0.05 & 1 \end{bmatrix} \begin{bmatrix} 1 & 10 \\ 0.05 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1.5 & 20 \\ 0.1 & 1.5 \end{bmatrix} \begin{bmatrix} 1 & 10 \\ 0.05 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2.5 & 35 \\ 0.175 & 2.5 \end{bmatrix} \end{aligned}$$

Step 6 of 12

Write the values $ABCD$ parameters of the cascaded network.

$$A_o = 2.5$$

$$B_o = 35 \Omega$$

$$C_o = 0.175 \text{ S}$$

$$D_o = 2.5$$

Step 7 of 12

Calculate the determinant of the matrix T , Δ_T .

$$\begin{aligned} \Delta_T &= (2.5)(2.5) - (0.175)(35) \\ &= 6.25 - 6.125 \\ &= 0.125 \end{aligned}$$

Step 8 of 12

Use the relationship between the z and the $ABCD$ parameters to find the z parameters of the overall cascaded network.

Calculate z_{11} .

$$z_{11} = \frac{A_o}{C_o}$$

Substitute 2.5 for A_o and 0.175 S for C_o .

$$\begin{aligned} z_{11} &= \frac{2.5}{0.175} \\ &= 14.28 \Omega \end{aligned}$$

Step 9 of 12

Calculate z_{12} .

$$z_{12} = \frac{\Delta_T}{C_o}$$

Substitute 0.125 for Δ_T and 0.175 S for C_o .

$$\begin{aligned} z_{12} &= \frac{0.125}{0.175} \\ &= 0.714 \Omega \end{aligned}$$

Step 10 of 12

Calculate z_{21} .

$$z_{21} = \frac{1}{C_o}$$

Substitute 0.175 S for C_o .

$$\begin{aligned} z_{21} &= \frac{1}{0.175} \\ &= 5.714 \Omega \end{aligned}$$

Step 11 of 12

Calculate z_{22} .

$$z_{22} = \frac{D_o}{C_o}$$

Substitute 2.5 for A_o and 0.175 S for C_o .

$$\begin{aligned} z_{22} &= \frac{2.5}{0.175} \\ &= 14.28 \Omega \end{aligned}$$

Step 12 of 12

Write the z parameters in the matrix form.

$$z = \begin{bmatrix} 14.28 & 0.714 \\ 5.714 & 14.28 \end{bmatrix} \Omega$$

Therefore the z parameters are $\boxed{\begin{bmatrix} 14.28 & 0.714 \\ 5.714 & 14.28 \end{bmatrix} \Omega}$.